



JINGTAO TANG (汤景韬)

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Google Scholar

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EDUCATION

Ph.D. Candidate

Computing Science

@ SFU Jan. '23 – Present

M.Eng.

Software Engineering

@ ECNU Sep. '18 – Jun. '21

B.Eng.

Software Engineering

@ ECNU Sep. '14 – Jun. '18

RESEARCH INTERESTS

Multi-Robot Motion Planning

Coverage Path Planning

Graphs of Convex Sets (GCS)

Heuristic Search

Combinatorial Optimization

AWARDS

Clark Wilson Graduate Scholarship, 2026

Best Paper Award, RSS Workshop on Scalable and Resilient Multi-Robot Systems (MRS), 2025

SFU-FAS Graduate Fellowship, 2025

Pacific Metals / Leon Lotzkar Memorial Graduate Scholarship, 2025

Robar Industries Ltd. Graduate Scholarship, 2024

Outstanding Graduate, Shanghai and ECNU, 2021

National Scholarship, China, 2020

ABOUT ME

I am a Ph.D. candidate in the *AIRob Lab* at Simon Fraser University, advised by Professor *Hang Ma*. I also collaborate with Professor *Richard Zhang* at *Augmenta*. Previously, I worked at the *Shenzhen Institute of Artificial Intelligence and Robotics for Society*, advised by Professor *Tin Lun Lam*, and studied in the *Intelligent Motion Planning and Vision Lab* at East China Normal University (ECNU), advised by Professor *Xinyu Zhang*. My research centers on multi-robot motion and coverage planning, with a focus on mixed discrete/continuous optimization on Graphs of Convex Sets (GCS) and scalable Multi-Robot Coverage Path Planning (MCP). I am also interested in deployment-oriented planning for real-world robotic and autonomous building-design systems.

EXPERIENCE

Routing Conduit Systems for Autonomous Building Design

Research Intern

Dec. 2025 – Present

Augmenta

I research methods to accelerate Augmenta's conduit-routing pipeline for large-scale buildings. The task is to plan multiple conduit routes from sources to targets in 3D routable space while satisfying cost and design constraints.

Mixed Discrete/Continuous Planning on Graphs of Convex Sets (GCS)

Ph.D. Candidate

Dec. 2024 – Present

SFU AIRob Lab

This topic involves planning on GCS — a powerful representation that decomposes the configuration space into convex regions connected by a graph. GCS naturally renders classic graph optimization problems into mixed discrete/continuous problems:

- Multi-Agent Pathfinding (MAPF) to Multi-Robot Motion Planning (MRMP):** We propose Space-Time GCS (ST-GCS), which systematically covers space-time collision-free regions with convex sets rather than random sampling. ST-GCS extends GCS into the time dimension, formulating time-optimal trajectories through unified convex optimization with velocity bounds and flexible arrival times.
- Traveling Salesman Problem (TSP) to GCS-TSP:** We developed a hierarchical framework optimally solving GCS-TSP where edge costs depend on trajectory selection through convex regions. It combines combinatorial tour search with convex trajectory optimization. We demonstrated on various GCS-TSP applications, including coverage and inspection planning, and task and motion planning.

Multi-Robot Coverage Path Planning (MCP)

Ph.D. Candidate

Jan. 2023 – Nov. 2024

SFU AIRob Lab

MCP plans for multiple robots to jointly cover a given workspace. This topic encompasses the following projects via combinatorial optimization and heuristic search:

- Graph-Based Formulation:** We formulated MCP as a min-max tree cover problem using mixed-integer linear programming on a grid graph representation of the workspace. While optimal, it scales only to 32×32 grids with 12 robots.
- Large-Scale MCP with Local Search:** We developed an efficient framework scaling to 256×256 grids with 100 robots, featuring a bounded-suboptimal single-robot CPP algorithm and adaptive spatiotemporal deconflicting for real-world deployment.
- Global MCP with Space-Filling Curves:** We extended connected *Fermat* spiral algorithms from 3D printing into MCP, generating smooth, continuous coverage paths for arbitrarily-shaped workspaces without grid decomposition.

SERVICES

Conference Reviewer: ICRA, IROS, AAAI, ECAI, ICAPS

Journal Reviewer: IEEE T-RO, RA-L, RAS, EAAI

Organizer: NWRS-25 (2025)

Volunteer: AAAI-24 (2024); ROS Summer School China (2018–2020)

Planning for Decentralized Heterogeneous Multi-Robot Systems (HMRS)

🌐 Research Assistant

📅 Jul. 2021 – Dec. 2022

📍 AIRS/CUHKSZ

This topic covers two Deep Reinforcement Learning (DRL) research projects:

1. **Coverage Planning:** We developed an RL solution for energy-limited worker robots and unlimited-energy station robots, where the workers are dedicated to coverage work and the stations recharge the workers. We trained the system with intrinsic curiosity and curriculum learning to address sparse rewards. The system was successfully deployed in real-world multi-robot applications.
2. **Adaptive Formation Control:** We considered the problem of coordinating a decentralized HMRS of quadrotors for adaptive formation control, where the formation evolves throughout the task. We formulated it as goal-conditioned RL, achieving rapid training in minutes using IsaacGym's GPU-accelerated parallel processing.

Robot Swarms for Large-scale Ecological Restoration Task

🌐 M.Eng. Student

📅 Jun. 2019 – Apr. 2021

📍 ECNU

In this project, we aimed to deploy multiple robots for the ecological restoration task in deserts. We abstracted the task into two research problems: 1) multi-robot coverage path planning with energy constraints; 2) generating feasible terrain-adaptive trajectories for wheeled robots. The system was successfully deployed in two deserts in China.

Projection Formats for Encoding 360° Virtual Reality (VR) Videos

🌐 M.Eng. Student

📅 Aug. 2018 – May. 2019

📍 ECNU

In this project, we aimed to improve the user experiences of telepresence for VR headsets. The main idea is to design efficient formats to encode/decode panoramic VR videos as regular rectangular videos for storage and content distribution.

PUBLICATIONS

- GHOST: Solving the Traveling Salesman Problem on Graphs of Convex Sets | **Jingtao Tang**, Hang Ma | AAAI-26, 2026 | [🌐 Homepage](#) | [🔗 Code](#)
- Space-Time Graphs of Convex Sets for Multi-Robot Motion Planning | **Jingtao Tang**, Zining Mao, Lufan Yang, Hang Ma | IROS-25, 2025; RSS 2025 Workshop on Scalable and Resilient Multi-Robot Systems (**Best Paper Award**) | [🌐 Homepage](#) | [🔗 Code](#)
- Large-Scale Multi-Robot Coverage Path Planning on Grids with Path Deconfliction | **Jingtao Tang**, Zining Mao, Hang Ma | IEEE Transactions on Robotics (T-RO), 41:3348–3367, 2025 | [🌐 Homepage](#) | [🔗 Code](#)
- Multi-Robot Connected Fermat Spiral Coverage | **Jingtao Tang**, Hang Ma | ICAPS-24, 2024 | [🔗 Code](#)
- Large-Scale Multi-Robot Coverage Path Planning via Local Search | **Jingtao Tang**, Hang Ma | AAAI-24, 2024 | [🌐 Homepage](#) | [🔗 Code](#)
- Mixed-Integer Programming for Time-Optimal Multi-Robot Coverage Path Planning with Efficient Heuristics | **Jingtao Tang**, Hang Ma | IEEE Robotics and Automation Letters (RA-L), 8(10):6491–6498, 2023 | [🔗 Code](#)
- TMSTC*: A Turn-Minimizing Algorithm for Multi-Robot Coverage Path Planning | Junjie Lu, Bi Zeng, **Jingtao Tang**, Tin Lun Lam, Junbin Wen | IEEE Robotics and Automation Letters (RA-L), 8(8):5275–5282, 2023 | [🔗 Code](#)
- Learning to Coordinate for a Worker-Station Multi-Robot System in Planar Coverage Tasks | **Jingtao Tang**, Yuan Gao, Tin Lun Lam | IEEE Robotics and Automation Letters (RA-L), 7(4):12315–12322, 2022
- MSTC*: Multi-Robot Coverage Path Planning under Physical Constraint | **Jingtao Tang**, Chun Sun, Xinyu Zhang | ICRA-21, 2021 | [🔗 Code](#)
- A Hybrid Projection Format for Encoding 360 VR Videos | **Jingtao Tang**, Xinyu Zhang | IEEE VR, 2019